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Laporan Praktikum 14

➤ 13.2 Persiapan

➤ Kode Program Customer7.java;

```
package Praktikum14;
public class Customer7 {
    public int id;
    public String name;

    public Customer7() {
    }

    public Customer7(int id, String name) {
        this.id = id;
        this.name = name;
    }

    public String toString() {
        return " ID: " + this.id + " Name: " + this.name;
    }
}
```

➤ Kode Program Book7.java;

```
package Praktikum14;
public class Book7 {
    public String isbn;
    public String title;

    public Book7() {
    }

    public Book7(String isbn, String title) {
        this.isbn = isbn;
        this.title = title;
    }

    public String toString() {
        return " ISBN: " + this.isbn + " Title: " + this.title;
    }
}
```

➤ 13.3 Praktikum - Implementasi ArrayList

➤ Kode Program DemoArrayList7.java;

```
package Praktikum14;
```

```

import java.util.ArrayList;
public class DemoArrayList7 {
    public static void main(String[] args) {
        ArrayList<Customer7> customers = new ArrayList<>();

        Customer7 customer1 = new Customer7(1, "Zakia");
        Customer7 customer2 = new Customer7(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new Customer7(4, "Cica"));

        customers.add(2, new Customer7(100, "Rosa"));

        System.out.println(customers.indexOf(customer2));

        Customer7 customer = customers.get(1);
        System.out.println(customer.name);
        customer.name = "Budi Utomo";

        ArrayList<Customer7> newCustomers = new ArrayList<>();
        newCustomers.add(new Customer7(201, "Della"));
        newCustomers.add(new Customer7(202, "Victor"));
        newCustomers.add(new Customer7(203, "Sarah"));

        customers.addAll(newCustomers);

        for (Customer7 cust : customers) {
            System.out.println(cust.toString());
        }
    }
}

```

5. Compile dan run kode program, di mana object yang baru ditambahkan? Di awal, di tengah, atau di akhir collection?

```
customers.add(new Customer7(id: 4, name: "Cica"));
```

```

ID: 1 Name: Zakia
ID: 5 Name: Budi
ID: 4 Name: Cica

```

Jawab: Di akhir collection

7. Compile dan run kode program. Index pada ArrayList dimulai dari 0 atau 1?

```
customers.add(index: 2, new Customer7(id: 100, name: "Rosa"));
```

```

ID: 1 Name: Zakia
ID: 5 Name: Budi
ID: 100 Name: Rosa
ID: 4 Name: Cica

```

Jawab: Dimulai dari indeks ke-0

10. Cobalah hapus angka 2 saat instansiasi object customers. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

```
ArrayList<Customer7> customers = new ArrayList<>();
```

```
1
Budi
  ID: 1 Name: Zakia
  ID: 5 Name: Budi Utomo
  ID: 100 Name: Rosa
  ID: 4 Name: Cica
```

Jawab: Ya, ArrayList dapat diinisialisasi tanpa menentukan size di awal.

11. Anda juga dapat menambahkan sekumpulan customer baru ke dalam ArrayList secara sekaligus. Misalnya terdapat ArrayList newCustomers. Tambahkan seluruh object customer sekaligus ke dalam customers.

```
ArrayList<Customer> newCustomers = new ArrayList<>();
newCustomers.add(new Customer(201, "Della"));
newCustomers.add(new Customer(202, "Victor"));
newCustomers.add(new Customer(203, "Sarah"));

customers.addAll(newCustomers);

for (Customer cust : customers) {
    System.out.println(cust.toString());
}
```

```
1
Budi
  ID: 1 Name: Zakia
  ID: 5 Name: Budi Utomo
  ID: 100 Name: Rosa
  ID: 4 Name: Cica
  ID: 201 Name: Della
  ID: 202 Name: Victor
  ID: 203 Name: Sarah
```

Jawab:

➤ 13.4 Praktikum - Implementasi Stack

➤ Kode Program StackDemo7.java;

```
package Praktikum14;
import java.util.Stack;
public class StackDemo7 {
    public static void main(String[] args) {
        Book7 book1 = new Book7("1234", "Dasar pemrograman");
```

```

Book7 book2 = new Book7("7145", "Hafalah Shalat Delisa");
Book7 book3 = new Book7("3562", "Muhammad Al-Fatih");

Stack<Book7> books = new Stack<>();
books.push(book1);
books.push(book2);
books.push(book3);

Book7 temp = books.peek();

if (temp != null) {
    System.out.println(temp.toString());
}

Book7 temp2 = books.pop();
if (temp2 != null) {
    System.out.println(temp2.toString());
}

for (Book7 book : books) {
    System.out.println(book.toString());
}

int posisi = books.search(book1);
System.out.println("Buku ini berada di posisi " + posisi);
}
}

```

5. Mengapa perlu ada pengecekan (temp != null)?

Jawab: Pengecekan temp != null diperlukan untuk mencegah program error ketika objek kosong, sehingga toString() hanya dijalankan jika variabel temp benar-benar memiliki data.

6. Lakukan looping untuk mencetak data buku pada stack

```

ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar pemrograman

```

Jawab: ISBN: 7145 Title: Hafalah Shalat Delisa

8. Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

Jawab:

```

int posisi = books.search(book1);
System.out.println("Buku ini berada di posisi " + posisi);

```

```

ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar pemrograman
ISBN: 7145 Title: Hafalah Shalat Delisa

```

Buku ini berada di posisi 2

➤ **13.5 Praktikum - Implementasi TreeSet**

➤ **Kode Program TreeSetDemo7.java;**

```
package Praktikum14;
import java.util.TreeSet;
public class TreeSetDemo7 {
    public static void main(String[] args) {
        TreeSet<String> fruits = new TreeSet<>();

        fruits.add("Mangga");
        fruits.add("Apel");
        fruits.add("Jeruk");
        fruits.add("Jambu");

        for (String temp : fruits) {
            System.out.println(temp);
        }

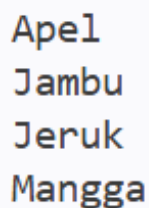
        System.out.println("First: " + fruits.first());
        System.out.println("Last: " + fruits.last());

        fruits.remove("Jeruk");
        System.out.println("Setelah remove " + fruits);

        fruits.pollFirst();
        System.out.println("Setelah poll first " + fruits);

        fruits.pollLast();
        System.out.println("Setelah poll last " + fruits);
    }
}
```

4. Compile dan run program. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?



```
Apel
Jambu
Jeruk
Mangga
```

Jawab: Urutan output berbeda karena TreeSet tidak menyimpan elemen berdasarkan urutan input, melainkan mengurutkannya secara otomatis menurut urutan alami (String → alfabetis).

6. Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?

Jawab:

- first(): Mengambil **elemen pertama (terkecil)** berdasarkan urutan sorting, **tanpa menghapusnya**
- last(): Mengambil **elemen terakhir (terbesar)** berdasarkan urutan sorting, **tanpa menghapusnya**.
- remove(): Menghapus elemen tertentu yang nilainya sama dengan objek yang diberikan.
- pollFirst(): Mengambil **dan menghapus** elemen pertama (terkecil).
- pollLast(): Mengambil **dan menghapus** elemen terakhir (terbesar).

➤ **Hasil Output:**

```
Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah remove [Apel, Jambu, Mangga]
Setelah poll first [Jambu, Mangga]
Setelah poll last [Jambu]
```

➤ **13.6 Praktikum – Sorting**

➤ **Modifikasi DemoArrayList7.java;**

```
package Praktikum14;
import java.util.ArrayList;
import java.util.Collections;
public class DemoArrayList7 {
    public static void main(String[] args) {
        ArrayList<Customer7> customers = new ArrayList<>();

        Customer7 customer1 = new Customer7(1, "Zakia");
        Customer7 customer2 = new Customer7(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new Customer7(4, "Cica"));

        customers.add(2, new Customer7(100, "Rosa"));

        System.out.println(customers.indexOf(customer2));

        Customer7 customer = customers.get(1);
        System.out.println(customer.name);
        customer.name = "Budi Utomo";
    }
}
```

```

        ArrayList<Customer7> newCustomers = new ArrayList<>();
        newCustomers.add(new Customer7(201, "Della"));
        newCustomers.add(new Customer7(202, "Victor"));
        newCustomers.add(new Customer7(203, "Sarah"));

        customers.addAll(newCustomers);

        for (Customer7 cust : customers) {
            System.out.println(cust.toString());
        }

        ArrayList<String> daftarSiswa = new ArrayList<>();
        daftarSiswa.add("Zainab");
        daftarSiswa.add("Andi");
        daftarSiswa.add("Rara");
        Collections.sort(daftarSiswa);

        System.out.println(daftarSiswa);

        customers.sort((c1, c2) -> c1.name.compareTo(c2.name));
    }
}

```

➤ Hasil Output

```

1
Budi
ID: 1 Name: Zakia
ID: 5 Name: Budi Utomo
ID: 100 Name: Rosa
ID: 4 Name: Cica
ID: 201 Name: Della
ID: 202 Name: Victor
ID: 203 Name: Sarah
[Andi, Rara, Zainab]

```